

6 Many predatory animals produce toxins to disable their prey. Some of these toxins inhibit nervous communication.

(a) Nerve impulses are transmitted along axons. Some axons are myelinated.

(i) Which type of cell produces myelin?

(1)

- ☐ A neurone
- ☐ B rod cell
- ☐ C Schwann cell
- ☐ D lymphocyte

(ii) Which row correctly compares the conduction of an impulse along a myelinated neurone with conduction along a non-myelinated neurone?

(1)

	Compared with a non-myelinated neurone the speed of conduction in a myelinated neurone is	Because membrane depolarisation only takes place in the axon membrane
☐ A	faster	at nodes of Ranvier
☐ B	faster	in between nodes of Ranvier
☐ C	slower	at nodes of Ranvier
☐ D	slower	in between nodes of Ranvier

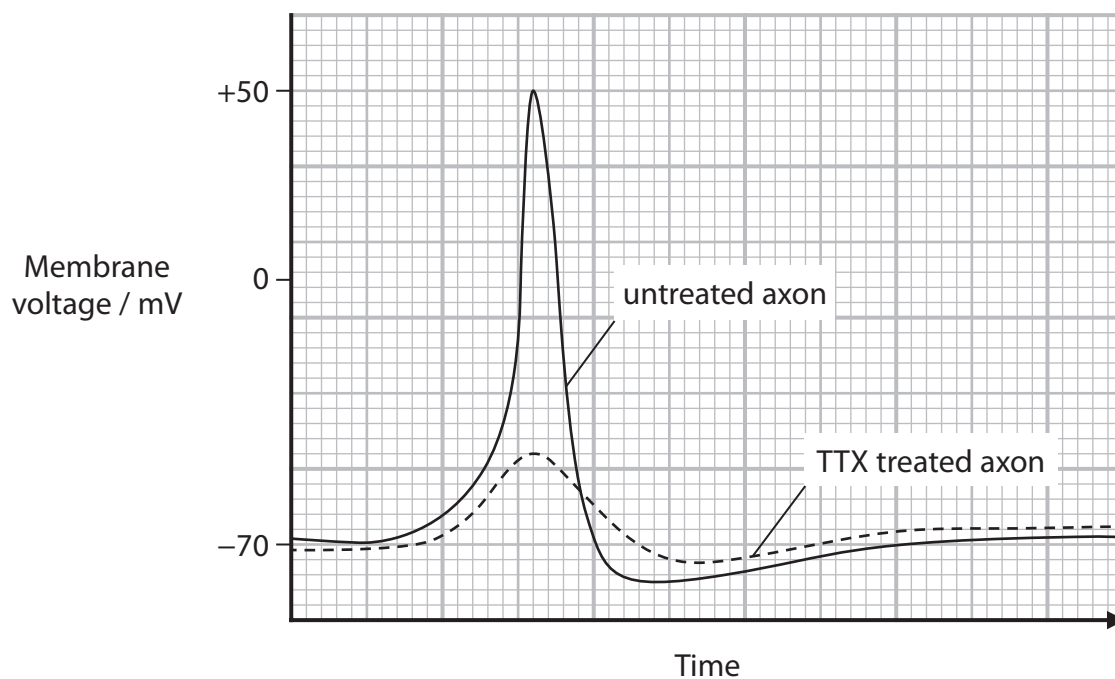
(b) Describe the role of ion transport in maintaining the resting potential of a neurone.

(4)

This image shows a full page of white paper with horizontal dashed lines, typical of primary school writing paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings on the paper.

(c) The pufferfish produces a neurotoxin called tetrodotoxin (TTX).

The graph shows the effect of TTX on the nerve impulse in an axon.



Deduce how TTX inhibits nerve impulses in an axon.

(3)

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(Total for Question 6 = 9 marks)